

When Moving Target Defense Meets Attack Prediction in Digital Twins: A Convolutional and Hierarchical Reinforcement Learning Approach

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Abstract—With rapid development of emerging technologies for Internet of Things (IoT), digital twins (DT) have been proposed to support a wide variety of applications. A mobile network is expected to be integrated with DT to form a DT mobile network (DTMN). Unfortunately, DTMN still faces security threats, which have attracted great research attention. Current defense mechanisms are mostly static, i.e., responding after attacks happening. To solve the aforementioned problem, moving target defense (MTD) has been proposed as an innovative solution. However, there exist three major challenges when applying MTD into DTMN. Firstly, less emphasis was paid to collaborative scheduling between multiple MTD schemes, which can improve the security of DTMN. Secondly, MTD schemes require lots of network resources, but few works focus on the time allocation of multiple MTD schemes to reduce network resource consumption. Thirdly, existing defense strategies only rely on

current information, but do not consider future information. In this paper, we propose a collaborative mutation-based MTD (CM-MTD) in DTMN. We mainly consider two MTD schemes called host address mutation (HAM) and route mutation (RM), respectively, which adjust network properties and invalidate different stages of cyber kill chain. We firstly formulate a semi-Markov decision process (SMDP) to model time-varying security events and dynamic deployment of multiple MTD schemes. Then, security events are predicted by long short-term memory (LSTM), which are regarded as network states in SMDP. Next, infeasible actions that do not satisfy network constraints will be removed from the action space of the SMDP. Lastly, we design a hierarchical deep reinforcement learning algorithm for collaborative scheduling. Simulation results highlight the effectiveness of CM-MTD compared with baseline solutions.

Index Terms—Digital twins, moving target defense, attack prediction, hierarchical deep reinforcement learning.

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I. INTRODUCTION

ACCORDING to the conclusion in [1], there will be 24 billion physical devices connected to Internet by 2050, such as wearable devices, mobile phones, and electric vehicles. The vast number of diverse physical devices connect with each other by using heterogeneous (wired/wireless) communications, which are regarded as Internet of Things (IoT) [2]. As one of the most important network scenarios in IoT, mobile network has become extensive existence for many years. On the other hand, emerging technologies (e.g., artificial intelligence (AI) [3], 5G and beyond [4], and software-defined networking (SDN) [5]) in IoT pave the way for the rapid development of digital twins (DT). Based on the statistics from [6], the size of DT market was initially valued at \$3.1 billion in 2020, and will be expected to reach \$48.2 billion by 2026. DT [7] aims to characterize physical assets through digital assets, whose applications include manufacturing, aviation, healthcare, intelligent transportation systems, and smart cities. These DT applications have diverse and strict network requirements such as quality of service (QoS), and network latency. To satisfy the aforementioned network requirements, DT mobile network (DTMN) is proposed to combine DT and mobile networks, thereby improving the network performance and reducing the cost of communication and computation.

In general DT, there exist physical space, digital space and communication space that connect physical and digital spaces [8]. The communication space provides bidirectional

interfaces, i.e., information from physical assets is handled by digital assets, by which digital assets create new information that may be sent back to physical space. According to the taxonomy in [9], DTMN has two aspects, which are *DT for mobile networks* and *mobile networks for DT*. For the latter, security becomes a critical issue because the communication space is easy to be compromised by cyber threats, which will lead to serious damage for DTMN [6]. In recent years, great attention has focused on security issues in DTMN. For example, Gehrman et al. [10] designed a security architecture for industrial automation and control systems. The work in [11] introduced a secure autonomous system for smart manufacturing transportation. Akbarian et al. [12] introduced a DT-based security framework for industrial control systems, which consists of attack detection and mitigation. Although aforementioned defense mechanisms are designed by considering the network characteristic of DTMN, but mostly respond after attacks happening, and hence, the effectiveness of these methods is seriously limited.

As an innovative solution to solve the challenge of “reactive response”, moving target defense (MTD) was proposed as a game-changing technique [13], [14]. To defend against cyber threats, MTD schemes dynamically adjust corresponding network properties (e.g., transmission routes, and IP addresses) to invalidate the prior knowledge of adversary. However, there still exist three important problems when trying to apply MTD into DTMN. Firstly, various MTD schemes are designed by considering different attack types, but single MTD scheme cannot protect DTMN comprehensively. Currently, few works consider the collaborative scheduling of multiple MTD schemes. Secondly, although the continuous-time deployment of multiple MTD schemes in DTMN will improve the defense effectiveness, but will also cost a large amount of network resources. However, previous works rarely focus on the deployment time allocation of multiple MTD schemes to reduce resource consumption. Finally, existing MTD methods only make defense decisions by current information, but lacks considering the future results of attack-defense confrontation.

To address aforementioned problems, we propose a collaborative mutation-based MTD (CM-MTD) scheme in DTMN. In this paper, we mainly consider two representative MTD schemes called host address mutation (HAM) and route mutation (RM) respectively. These methods both adjust network properties, but destroy different stages of cyber kill chain (i.e., network reconnaissance and attack execution phases), which means that HAM and RM are suitable to collaborate with each other. CM-MTD dynamically schedules the deployment of HAM and RM based on attack prediction and hierarchical deep reinforcement learning (HDRL). Our previous works [15], [16], [17] have proved that deep reinforcement learning (DRL) is a promising way to improve the defense performance of MTD significantly. By adopting the long short-term memory (LSTM), CM-MTD firstly predicts what future malicious behaviors are for adversary. In recent years, LSTM has been regarded as a typical way to catch the short-term and long-term information and handle the time series data. Then, predicted security events will be the current network state of HDRL. In HDRL, the upper layer determines

the deployment category of MTD schemes while the lower layer decides mutated routes and/or assigned IP address spaces by learning from malicious behaviors. To the best of our knowledge, our work is the first contribution to design a collaborative and intelligent MTD scheme in DTMN.

To sum up, the main contributions of this paper are listed as follows:

- 1) We formulate a semi-Markov decision process (SMDP) model to describe time-varying security events and the dynamic deployment of multiple MTD schemes. Unlike traditional MDP, SMDP models the continuous-time discrete-event scenario, and can handle sequential decision-making. Without relying only on current information, we design network state as the jointly predicted security events of all network nodes because security events are directly related to reward and action to be taken. In addition, we define macro-action as choosing the deployment strategy, and action as deploying corresponding MTD schemes according to the selected macro-action.
- 2) Future security events are predicted by LSTM based on security logs, which provides the accurate prediction to formulate the network state. In addition, the problem of assigning IP address spaces and selecting mutated routes is formalized as a constrained satisfaction problem based on satisfiability modulo theory (SMT), which considers multiple practical network constraints including operation, flow table size, and QoS. By solving the aforementioned constrained satisfaction problem, infeasible actions will be removed from the action space of SMDP, making the solution is more efficiently to obtain.
- 3) Based on the definitions of SMDP, we design a HDRL algorithm for collaborative scheduling, where hierarchical structure can reduce the complexity of mutation actions. The macro-action in the upper layer decides the deployment category of HAM and/or RM according to predicted security events. Then, the lower layer periodically chooses mutated routes and/or assigned IP address spaces based on the selected macro-action. The objective of HDRL is to improve the defense performance and reduce resource overheads by learning from malicious behaviors with the adaptive deployment of HAM and RM.
- 4) A series of simulations in Mininet-WiFi are performed to validate the effectiveness of CM-MTD including prediction, defense, network and convergence performances. We also implemented a proof-of-concept prototype system to conduct experiments. As a conclusion, compared with baseline methods, our proposed CM-MTD can improve defense performance without the severe impact on QoS.

The rest of this paper is organized as follows: Section II summarizes the related work. In Section III, network, threat and SMDP models are explained respectively. System overview is introduced in Section IV. Section V formalizes the feasible actions of collaborative scheduling in DTMN.

In Section VI, attack prediction and our proposed HDRL algorithm are presented in detail. Section VII shows the evaluation of multiple performances. Lastly, Section VIII concludes this paper and draws future research.

II. RELATED WORK

Recently, great attention has been paid on DTMN applications in a large variety of fields including manufacturing [18], healthcare [19], autonomous vehicles [11], smart cities [20], and so on. However, whatever traditional IoT or emerging DTMN, they are both easy to become the attacked target for adversary [6]. Based on the concept of cyber kill chain, network reconnaissance and distributed denial of service (DDoS) are two main stages to breach security goals, which aim to degrade the network performance. In fact, some works have focused on network reconnaissance and DDoS in IoT. For example, to defend against DDoS, Cvitic et al. [22] designed a traffic detection method based on logistic trees for different IoT devices. The work in [23] proposed to detect malicious traffic from IoT devices based on the analysis of session IP counter and payload. Sudheera et al. [24] introduced ADEPT to detect and identify the attack stages of coordinated attacks. The work in [25] introduced an adaptively online learning method for the real-time detection of botnet. Meanwhile, DT security also needs to be protected from cyber threats. Gehrman et al. [10] designed a security architecture for industrial automation and control system in DT. The work in [11] introduced a secure autonomous system for smart manufacturing transportation. Akbarian et al. [12] presented a DT-based security framework for industrial control systems, which consists of attack detection and mitigation. However, almost all of aforementioned methods and other existing defense schemes are reactive, which means that network system has already been compromised for a period of time, causing significant damage before defense mechanisms begin executing.

Fortunately, MTD has been employed to handle this serious problem. Two MTD schemes called HAM and RM [13] were designed to defend against network reconnaissance and DDoS respectively. HAM mutates IP addresses periodically to increase the complexity, diversity, and randomness of end-hosts while RM mutates transmission routes periodically to avoid links and nodes compromised by DDoS. In field of HAM, Jafarian et al. [26] proposed to mutate IP addresses of end-hosts by using the hypothesis test to analyze and characterize adversarial behaviors. The work in [27] deployed decoy nodes with dynamic IP addresses, which aims to minimize the probability of being identified as honeypots. Sharma et al. [28], [29] proposed to convert real IP addresses to random and time-varying virtual IP addresses. In the field of RM, the work in [30] proposed to adjust the routes of multiple flows in SDN-based wireless sensor networks. Wang et al. [31] designed a RM scheme with adaptive mutation period. The work in [32] introduced an intelligent RM scheme based on DRL.

Although MTD solutions improve defense performance significantly, they cannot be applied into DTMN directly because

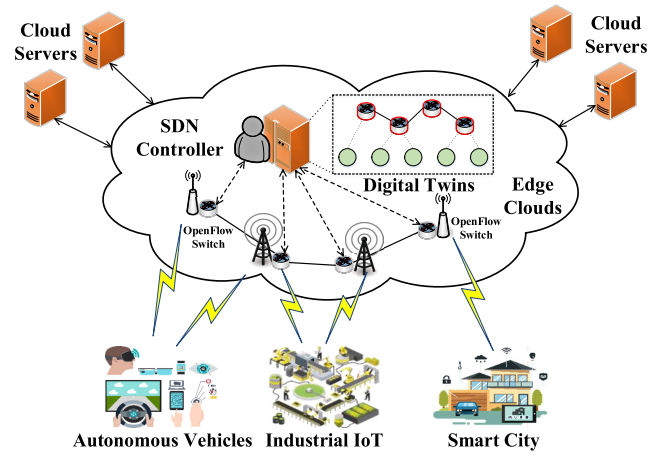


Fig. 1. The architecture of SDN-IoT in DTMN.

DTMN has a complex network scenario and diverse network requirements, which means that a single MTD scheme cannot protect DTMN comprehensively. **Actually, HAM disturbs the information collection of network reconnaissance while RM avoids links or nodes to be compromised by DDoS. As a result, they invalidate different stages of cyber kill chain. Therefore, HAM and RM can complement each other and yield stronger security protection.** Little prior work has been done to investigate the cooperation between multiple MTD schemes. In addition, the continuous deployment of HAM and RM will consume a large amount of network resource. To bridge these gaps, the present contribution proposes the CM-MTD scheme based on attack prediction and HDRL for the first time.

III. MODEL FORMULATION

In this section, we introduce network, threat and SMDP models respectively.

A. Network Model

Fig.1 illustrates the architecture of SDN-IoT in DTMN. The physical device layer consists of massively various sensors that support various applications (e.g., autonomous vehicles, industrial IoT, and smart city) and remote cloud servers that provide Internet service. Base stations (BSs) and road-side units (RSUs), where OpenFlow switches are located, act as access points for static or mobile devices (e.g., sensors and vehicles), and routers for cloud servers. The control layer consists of edge servers, where SDN controllers are located. The DT layer implements various applications including network visualization, security optimization, and so on.

The SDN-IoT in DTMN can be modeled as an undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{M}, \mathcal{E}\}$, where $\mathcal{V}$ is the set of IoT devices and cloud nodes v_i^n ($1 \leq i \leq n$), $\mathcal{M}$ is the set of OpenFlow switches v_j^s ($1 \leq j \leq m$), and $\mathcal{E}$ is the set of wireless/wired links connecting different nodes and/or switches. In the view of edge clouds, cloud servers, sensors and vehicles are both considered as endpoints.

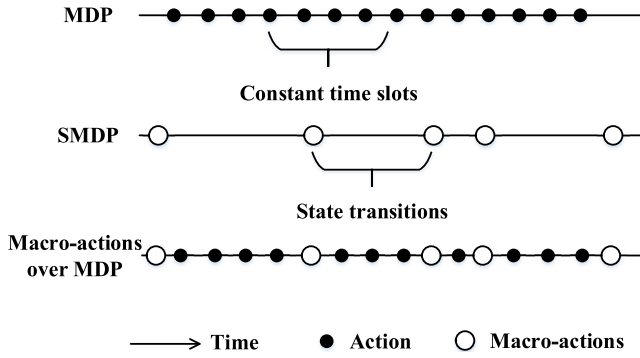


Fig. 2. The state trajectories of MDP and SMDP.

B. Threat Model

In DTMN, remote cloud servers provide Internet services for various DT applications, which is easy to become attacked targets. On the other hand, due to the resource-constrained characteristic of massive IoT devices, complex defense mechanisms cannot be deployed. Therefore, most IoT devices will be easily compromised by the adversary, and become the botnet. Then, the adversary launches cyber attacks to cloud servers by controlling the botnet, which will lead to network paralysis or even interruption.

In actual, based on cyber kill chain, the adversary usually adopts multi-steps attack including reconnaissance phase (e.g., IP and port scanning) and attack execution phase (e.g., DDoS flooding) [33], [34], [35]. In addition, we assume that the adversary is rational and has limited botnet resource to launch cyber attacks [36]. Usually, attack execution phase will last for a period of time (e.g., several hours or days), and then enter the next round of reconnaissance.

C. Semi-Markov Decision Process Model

Considering that collaborative scheduling has the characteristic of sequential decision-making, we formulate the dynamic deployment of multiple MTD schemes as an SMDP model, as shown in Fig.2. In the upper layer of SMDP, open circle represents the macro-action regarding whether HAM and/or RM are deployed, which will be decided firstly. In the lower layer of SMDP, time is divided into equal slots denoted as Δt . The solid black dot denotes selecting mutated routes and/or assigning IP address spaces based on macro-action. The upper and lower layers interact through reward feedback, where actions are executed subsequently when macro-actions change.

1) *Network State Space*: network state is defined as the security events of all network nodes in DTMN. We denote the network state as a vector $S_t = \{e_1^{t+1}, e_2^{t+1}, \dots, e_n^{t+1}\}$, where e_i^{t+1} is the predicted security event of network node v_i^n ($1 \leq i \leq n$) at time slot $t+1$. The future security events of network nodes are predicted based on LSTM, which will be introduced in Section VI-A. Assuming that the number of possible security events is L , the total number of network states will be $\Gamma = L^n$. Therefore, network state space is described as $\{S_1, \dots, S_\Gamma\}$.

2) *Action Space*: in this paper, the deployment of MTD schemes consists of two components: 1) assigning IP address spaces to network nodes; 2) selecting mutated routes for flows. Collaborative mutation is a joint action denoted as a multi-dimensional vector $A_t = \{\mathbb{I}_t, \mathbb{B}_t\}$, where $\mathbb{I}_t = \{I_{1,1}, \dots, I_{i,x}, \dots, I_{n,w}\}$ and $\mathbb{B}_t = \{B_1^{f_1}, \dots, B_m^{f_1}, \dots, B_1^{f_k}, \dots, B_m^{f_k}\}$. The value of $I_{i,x}$ ($1 \leq i \leq n, 1 \leq x \leq w$) is equal to 1 if x -th IP address space is assigned to node v_i^n , and otherwise $I_{i,x} = 0$. The value of $B_j^{f_y}$ ($1 \leq j \leq m, 1 \leq y \leq k$) is equal to 1 if OpenFlow switch v_j^s is selected into the route of flow f_y , and otherwise $B_j^{f_y} = 0$. We denote action space as $\mathcal{A}$ that consists of all possible actions. However, most actions in aforementioned space are infeasible because they do not satisfy multiple network constraints, which are formalized in Section V.

3) *Macro-Action Space*: macro-actions are considered as the generalization of actions. In this paper, macro-action space is defined as $\mathcal{O} = \{o_c, o_a, o_r, o_s\}$, where o_c denotes deploying both HAM and RM, o_a denotes only deploying HAM, o_r denotes only deploying RM, and o_s represents static IP addresses and routes. For any $o \in \mathcal{O}$, it is denoted as $\langle \mathcal{I}, \pi, \zeta \rangle$, where $\mathcal{I}$ is the initiation set, π is the action policy, and ζ is the termination condition. If $S_t \in \mathcal{I}$, macro-actions are available, and will be chosen based on policy π . In our network scenario, SDN controller can determine the macro-action at any network state, and adopt the determined policy. The termination condition ζ for any macro-action is after constant time slots.

4) *State Transitions*: based on the definition of network state, the changing of predicted security events will be regarded as the state transition.

5) *Reward Function*: rewards will be obtained when the action is determined. Reward has two parts: 1) defense reward R_d ; 2) resource consumption reward R_c . The objective of reward R_d is to evaluate the defense effectiveness, which is defined as follows:

$$R_d = \begin{cases} -\alpha_1 \sum_{i \in n} \Theta_{t,i} - \alpha_2 \Upsilon_t, & \text{if attacks are successful,} \\ \mathcal{C}, & \text{otherwise,} \end{cases} \quad (1)$$

where α_1 and α_2 are coefficients, $\Theta_{t,i}$ is the number of times that node v_i^n was scanned successfully at time slot t , Υ_t is the number of OpenFlow switches that are compromised by adversary at time slot t , and $\mathcal{C}$ is a positive constant. Reward R_c is used to evaluate the resource consumption caused by HAM and/or RM, which is defined as follows:

$$R_c = \begin{cases} W_t, & \text{if macro-action is one of } o_c, o_a \text{ and } o_r, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where W_t is the function of resource consumption defined as $W_t = -\gamma_1 \sum_{i \in n} e_i^a - \gamma_2 \sum_{x \in k} e_x^r$. Thereinto, γ_1 and γ_2 are coefficients, e_i^a denotes the resource consumption of HAM for network node v_i^n , and e_y^r denotes the resource consumption of RM for flow f_y . Correspondingly,

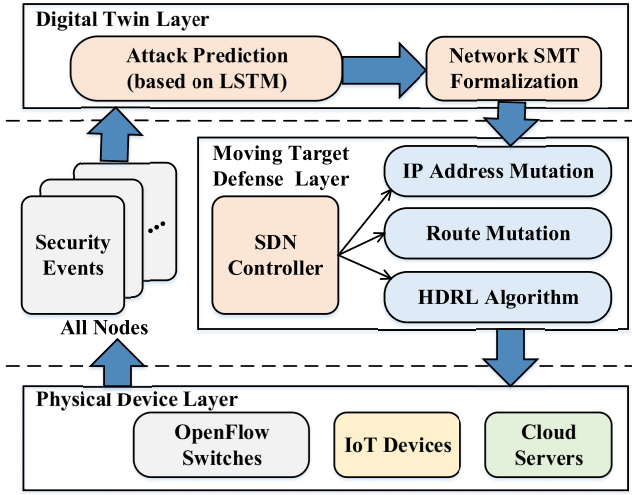


Fig. 3. The framework of our proposed CM-MTD scheme.

the total reward is denoted as:

$$R_{total} = R_d + R_c, \quad (3)$$

where R_{total} is calculated after each action, and the reward of macro-action is the sum of multiple rewards from chosen actions.

IV. SYSTEM OVERVIEW

To defend against multi-steps attack, we design CM-MTD scheme that adaptively adjusts the deployment time allocation of HAM and RM. HAM mutates the IP addresses of network nodes periodically to disturb reconnaissance from adversary. On the other hand, RM adjusts routes periodically to avoid OpenFlow switches that are compromised by DDoS, which aims to decrease the DDoS damage. The execution mechanisms of HAM and RM have been introduced in existing works [26], [27], [28], [29], [30], [31], [32], and hence, we omit their detail for brevity.

Fig.3 depicts the framework of our proposed CM-MTD scheme. In the physical device layer, security events are generated from the detection logs of network nodes including IoT devices and cloud servers. Various attack phases can be detected by existing security mechanisms (e.g., firewall and intrusion detection system). Then, these security events will be taken as the input of LSTM, which aims to predict the future security events of network nodes (details are given in Section VI-A). Next, infeasible actions that do not satisfy multiple network constraints will be removed from the action space of SMDP (details in Section V). Finally, the outputs of LSTM and SMT formalization will be regarded as the network state and action respectively in HDRL, which will learn the optimal policy of deploying HAM and/or RM by iteratively updating (details are given in Section VI-B).

V. COLLABORATIVE MUTATION FORMALIZATION

In this section, we formalize the problem of assigning IP address spaces and selecting mutated routes as a constrained satisfaction problem based on SMT. Binary variable b_i^x denotes

whether IP address space z_x is assigned to network node v_i^n . If so, $b_i^x = 1$, and otherwise $b_i^x = 0$. Binary variable d_j^y denotes whether OpenFlow switch v_j^s is selected into the route of flow f_y . If so, $d_j^y = 1$, and otherwise $d_j^y = 0$. Considering practical network conditions, the feasible actions of HAM and RM should satisfy multiple network constraints.

A. Operation Constraints

Collaborative mutation must satisfy basic network operation constraints. The constraints of assigning IP address spaces are formalized as follows:

$$\sum_{x=1}^w b_i^x \geq 1, \quad \forall v_i^n \in \mathcal{V}, \quad (4)$$

$$\sum_{i=1}^n b_i^x = 1, \quad \forall x \in [1, w], \quad (5)$$

Eq. (4) indicates that the number of IP address spaces assigned to same node is greater than or equal to one. Eq. (5) indicates that each IP address space must be assigned to only one node.

We suppose that each flow f_y ($1 \leq y \leq k$) has its source switch s_y and destination switch d_y . A valid mutated route of flow f_x with reachability constraints is formalized as follows:

$$\sum_{e_j \in \mathbb{O}_j} d_j^y = \sum_{e_j \in \mathbb{P}_j} d_j^y, \quad \forall v_j^s \in \mathcal{V}, \quad \forall f_y \in \mathcal{F}, \quad (6)$$

$$d_j^y \in \{0, 1\}, \quad \forall v_j^s \in \mathcal{V}, \quad \forall f_y \in \mathcal{F}, \quad (7)$$

where $\mathbb{O}_j$ denotes the set of outgoing flows of OpenFlow switch v_j^s , $\mathbb{P}_j$ denotes the set of incoming flows of OpenFlow switch v_j^s and $\mathcal{F}$ denotes the set of flows. Eq. (6) indicates that any OpenFlow switch in flow f_y must have the identical number of outgoing flows and incoming flows. Eq. (7) defines the value range of variable d_j^y .

B. Flow Table Size Constraint

When IP address spaces are assigned to different network nodes, the flow tables of OpenFlow switches must be updated to respond these changes. If adjacent IP address spaces are allocated to same network node, supernetting will be used to aggregate these spaces into larger ones, which can reduce the size of flow table. Assuming that binary variable $\mathcal{D}_{x_1, x_2}$ denotes whether IP address spaces z_{x_1} and z_{x_2} are nonadjacent. If so, $\mathcal{D}_{x_1, x_2} = 1$, and otherwise $\mathcal{D}_{x_1, x_2} = 0$. Then, binary variable $\mathcal{L}_i^{x_1, x_2}$ denotes whether nonadjacent IP address spaces z_{x_1} and z_{x_2} are assigned to same network node v_i^n . The value of $\mathcal{L}_i^{x_1, x_2}$ is calculated as follows:

$$\mathcal{L}_i^{x_1, x_2} = b_i^{x_1} \wedge b_i^{x_2} \wedge \mathcal{D}_{x_1, x_2}. \quad (8)$$

If more flows pass through OpenFlow switch v_i^s , the number of available entries will decrease in the flow table. When the size of flow table is large, OpenFlow switches cannot perform lookups timely. The constraint of the flow table size is formalized as:

$$\sum_{x_1=1}^m \sum_{x_2 \neq x_1} \sum_{i \in l_j^y} \mathcal{L}_i^{x_1, x_2} + \sum_{y=1}^k \sum_{j \in r_y} d_j^y \leq \mathbb{C}_j^{cap}, \quad \forall f_y \in \mathcal{F}, \quad (9)$$

where l_j^y denotes the set of network nodes connected to OpenFlow switch v_j^s , which belongs to the source or destination switch of flow f_y , r_y denotes the set of OpenFlow switches on the transmission route of flow f_y except source and destination switches and $\mathbb{C}_j^{cap}$ denotes the maximum capacity of the flow table for OpenFlow switch v_j^s .

C. Quality of Service Constraint

Collaborative mutation should satisfy the diverse QoS of DTMN, such as transmission delay and maximum hops. The QoS constraint is formalized as follows:

$$\sum_{j=1}^m d_j^y T_j^m + \sum_{j_1=1}^m \sum_{j_2 \neq j_1}^m d_{j_1}^y d_{j_2}^y T_l \leq \mathbb{C}_y^{del}, \quad (10)$$

where T_l denotes the link delay, $\mathbb{C}_y^{del}$ denotes the maximum delay threshold for flow f_y , and T_j^m will be the processing delay caused by modifying the IP address of packets if v_j^s is the source or destination switch. Otherwise, if v_j^s is the intermediate switch, T_j^m will be forward delay because of lookups.

VI. INTELLIGENT COLLABORATIVE MUTATION SCHEME

In this section, we firstly introduce that how to utilize LSTM for attack prediction. Then, we design a novel HDRL algorithm to decide the dynamic deployment of multiple MTD schemes.

A. Long Short-Term Memory for Attack Prediction

Attack prediction adopts the event sequences from historical collection to predict attack types and targets at future time slot. In recent years, LSTM has become a typical solution that handles the time series data [37], [38]. The output layer of LSTM will be the input layer of HDRL, which indicates that LSTM and HDRL can be integrated tightly. Few work considers LSTM to improve the macro-action efficiency of HDRL in the field of MTD. The architecture of cell in LSTM is defined as follows:

$$\begin{aligned} f_t &= \sigma(\omega_f \cdot [x_t, h_{t-1}] + b_f), \\ i_t &= \sigma(\omega_i \cdot [x_t, h_{t-1}] + b_i), \\ o_t &= \sigma(\omega_o \cdot [x_t, h_{t-1}] + b_o), \\ \widetilde{C}_t &= \tanh(\omega_c \cdot [x_t, h_{t-1}] + b_c), \\ \overline{C}_t &= f_t \odot \overline{C}_{t-1} + i_t \odot \widetilde{C}_t, \\ h_t &= o_t \odot \tanh(\overline{C}_t), \end{aligned} \quad (11)$$

where f_t is forget gate that decides the forgetting level of previous information h_{t-1} , i_t is input gate that decides how much the input x_t is collected by unit state $\overline{C}_t$, o_t is output gate that gives the output h_t . In addition, variables $\{\omega_f, \omega_i, \omega_o, \omega_c\}$ and $\{b_f, b_i, b_o, b_c\}$ denote the weight and bias of neural network respectively, and $\odot$ represents the Hadamard product.

Security events are generated from detection logs. The arrival of security event sequence for network node v_i^n is denoted as $E_i = \{e_i^1, e_i^2, \dots, e_i^t\}$. The security event arrival sequence of all network nodes is denoted as

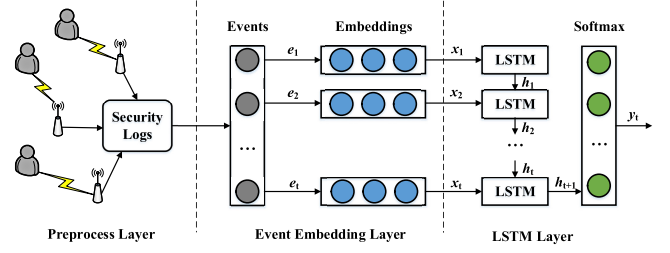


Fig. 4. Security logs are collected and extracted to events in preprocess layer, then put into event embedding layer to construct the context-based representation. Finally, LSTM is used to predict the future security event.

$\mathbb{E} = \{E_1, E_2, \dots, E_n\}^T$. Then, the neural network of attack prediction is built with $\phi(\mathbf{X}_p; \theta_p)$, where $\mathbf{X}_p$ denotes the input, and θ_p denotes the parameters of neural network. The training data of security event sequence is denoted as:

$$[X | Y] = \left[\begin{array}{cccc|c} E_k^1 & E_k^2 & \dots & E_k^L & E_k^{L+1} \\ E_k^2 & E_k^3 & \dots & E_k^{L+1} & E_k^{L+2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ E_k^{t-L} & E_k^{t-L+1} & \dots & E_k^{t-1} & E_k^t \end{array} \right], \quad (12)$$

where $k \in \{1, 2, \dots, n\}$, E_k^t denotes the security event collected by network node v_i^n at time slot t , $X_k^t = \{E_k^{t-L}, E_k^{t-L+1}, \dots, E_k^{t-1}\}$, $Y_k^t = E_k^t$, and each row is a sample data.

Fig. 4 illustrates the detailed architecture of attack prediction based on LSTM network (LSTMNet). In the preprocess layer, multiple features (e.g., timestamp, node ID, security event ID, and event description) are generated from security logs. In the event embedding layer, security events are mapped into the low dimensional space [33], [34], [39]. In the LSTM layer, LSTMNet will be formulated to obtain the high accuracy of attack prediction. Softmax function is added to get the probability distribution of future security events. Loss function adopts mean square error to directly measure the deviation between predicted and real values. Finally, an adaptive optimizer called Adam is used to minimize the loss function [40].

B. Hierarchical Deep Reinforcement Learning for Collaborative Mutation

In this subsection, we design a novel HDRL algorithm for collaborative mutation. As shown in Fig.5, the upper layer adopts DQN [41] to obtain the optimal policy of deployment decisions, and the lower layer adopts PPO [42] to learn optimal policy of selecting mutation actions. Network state will be output from LSTMNet, which predicts the future security events. In actual, the number of collectively predicted network nodes can be determined dynamically according to practical situation. This is because that when less network nodes are predicted separately, LSTMNet has low complexity. On the other hand, when more network nodes are predicted collectively, LSTMNet has better global accuracy.

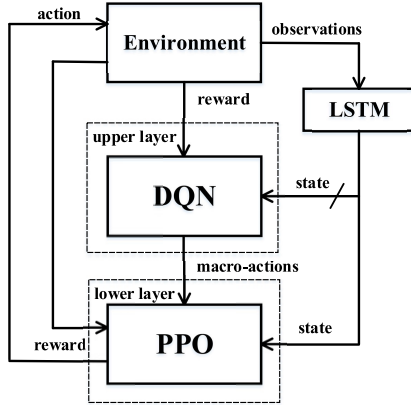


Fig. 5. The flowchart of our proposed HDRL algorithm.

Next, we introduce the overview of HDRL. Specifically, DQN estimates the following value function:

$$V_u^*(S_t, \mathcal{O}_t) = \max_{\pi_u} \mathbf{E} [R_t + \sum_{k=1}^{\infty} \gamma^k R_{t+k}], \quad (13)$$

where $\mathcal{O}_t$ denotes the macro-action at state S_t , and $\pi_u = P(\mathcal{O}|S)$ is the policy of choosing macro-actions. In DQN, the value function is approximated by deep neural network (DNN), which aims to minimize the loss function as follows:

$$\mathcal{L}(\theta_t) = \mathbf{E} [(y_t - Q_t(S_t, \mathcal{O}_t, \theta_t))^2], \quad (14)$$

where Q_t denotes the state-action value at time slot t , θ_t denotes the parameter of DNN at time slot t , y_t is the target value defined as $y_t = R_t + \max_{A'} Q(S', \mathcal{O}', \theta_{t-n})$, and θ_{t-n} is the value of parameter θ before n time slots. Similarly, for PPO, we have:

$$V_l^*(S_t, A_t; \mathcal{O}_t) = \max_{\pi_l} \mathbf{E} [R_t + \sum_{k=1}^N \gamma^k R_{t+k}], \quad (15)$$

where $\pi_l = P(A|S, \mathcal{O})$ is the policy of choosing mutation actions, and N denotes the number of time slots until moving into next macro-action. In PPO, parameterized policy is approximated by DNNs, and loss function is defined as follows:

$$\mathcal{L}^{clip}(\theta_t) = \mathbf{E} [\min(r_t(\theta_t), \text{clip}(r_t(\theta_t), 1 - \epsilon, 1 + \epsilon)) \hat{A}_t], \quad (16)$$

where $\hat{A}_t$ denotes the estimator function at time slot t , ϵ is the parameter to control the clip range and $r_t(\theta_t)$ denotes the policy probability ratio defined as:

$$r_t(\theta_t) = \frac{\pi(A_t|S_t, \mathcal{O}_t; \theta_t)}{\pi(A_t|S_t, \mathcal{O}_t; \theta_t^{old})}. \quad (17)$$

In addition, the objective of clip function is to constrain the value range of r_t .

The pseudo-code of our designed HDRL for collaborative mutation is shown in Algorithm 1. Parameters, the replay buffer and DNNs are initialized firstly (lines 1-8). In DQN, ϵ denotes greedy factor and α denotes the learning rate. In PPO, ξ is used to balance the tradeoff between variance and bias, ϵ is the parameter to control the clip range, and γ is the

Algorithm 1 Hierarchical Deep Reinforcement Learning for Collaborative Mutation

```

1: Set parameters  $\epsilon$  and  $\alpha$  in DQN;
2: Set parameters  $\xi$ ,  $\epsilon$ , and  $\gamma$  in PPO;
3: Set batch size  $T$  and minibatch size  $K$ ;
4: Initialize experience replay buffer  $\mathcal{B}_1 = \emptyset$ ,  $\mathcal{B}_2 = \emptyset$ ;
5: Initialize main Q-network with weight  $\theta_t^u$ ;
6: Initialize target Q-network with weight  $\theta_{t-n}^u = \theta_t^u$ ;
7: Initialize critic network  $V(S_k, \phi)$ ;
8: Initialize actor network with weight  $\theta_k^l$ ;
9: for  $episode = 1, 2, \dots, M$  do
10:   for  $t = 1, 2, \dots, T$  do
11:     Obtain current state  $S_t$  by LSTMNet;
12:     Generate a random probability  $p$ ;
13:     Select macro-action  $\mathcal{O}_t$  as,
14:     if  $p \leq \epsilon$  then
15:       Select a macro-action  $\mathcal{O}_t$  randomly;
16:     else
17:        $\mathcal{O}_t = \arg \max_{\mathcal{O}'} Q_t(S_t, \mathcal{O}_t; \theta_t^u)$ ;
18:     end if
19:     for  $k = 1, 2, \dots, K$  do
20:       Run policy  $\pi_{\theta_{k-n}^l}$  to select action  $A_k$ ;
21:       Deploy corresponding MTD schemes;
22:       Observe lower layer reward  $R_k^l$ ;
23:       Collect  $\mathcal{D}_k = (S_k, A_k, R_k^l, S_{k+1})$ ;
24:       Store into the replay buffer  $\mathcal{B}_2 = \mathcal{B}_2 \cup \mathcal{D}_k$ ;
25:       Calculate TD errors  $\delta_t$ ;
26:        $\hat{A}_k = \sum_{q=k}^K (\gamma \xi)^{q-k} \delta_q$ ;
27:       Estimate value target  $\hat{V}_k = \hat{A}_k + V(S_k, \phi)$ ;
28:     end for
29:     Observe upper layer reward  $R_t^u$ ;
30:     Obtain next state  $S_{t+1}$  by LSTMNet;
31:     Collect  $\mathcal{D}'_t = \{S_t, \mathcal{O}_t, R_t^u, S_{t+1}\}$ ;
32:     Store into the replay buffer  $\mathcal{B}_1 = \mathcal{B}_1 \cup \mathcal{D}'_t$ ;
33:   end for
34:   for  $epoch = 1, 2, \dots, U$  do
35:     Calculate the target Q-value  $y_t$ ;
36:     Minimize the loss function by gradient descent;
37:      $\mathcal{J}^{actor}(\theta) = \frac{1}{K} \sum_{k=1}^K \min(r_k(\theta), \text{clip}(r_k(\theta),$ 
38:        $1 - \epsilon, 1 + \epsilon)) \hat{A}_k$ ;
39:     Update  $\theta_k^l$  by  $\nabla_{\theta} \mathcal{J}^{actor}$ ;
40:      $\mathcal{J}^{critic}(\theta) = -\frac{1}{K} \sum_{k=1}^K (\hat{V}_k - V(S_k, \phi))^2$ ;
41:     Update  $\phi$  by  $\nabla_{\phi} \mathcal{J}^{critic}$ ;
42:   end for
43:    $\theta_{t-n}^u \leftarrow \theta_t^u$ ;
44:    $\theta_{k-n}^l \leftarrow \theta_k^l$ ;
45: end for

```

discount factor between 0 and 1. Algorithm 1 mainly has two loop parts (line 9). The first part describes that how to obtain state-action samples iteratively (lines 10-33). On each time slot, the network state will be obtained by LSTMNet (line 11). Then, macro-action will be chosen based on the ϵ -greedy strategy (lines 12-18). After selecting a macro-action, mutation actions will be decided by PPO in the lower layer (lines 19-28). The SDN controller will adopt policy $\pi_{\theta_{k-n}^l}$ to

choose an action that deploys corresponding MTD schemes, and then observes the reward of lower layer influenced by cyber attacks (lines 20-22). The state-action sample is stored into replay buffer $\mathcal{B}_2$. Next, generalized advantage estimation approach is used to estimate advantage and value functions respectively (lines 24-27). When finishing K iterations in the lower layer, the SDN controller will obtain the reward of upper layer and next network state by LSTMNet (lines 29-30). The sample of state and macro-action is stored into replay buffer $\mathcal{B}_1$ (lines 31-32). The second part describes that how to learn from samples. In DQN, the loss function is minimized by gradient descent method (lines 35-36). In PPO, the policy gradient and value function are calculated respectively. In addition, various parameters are updated correspondingly (lines 37-40). Finally, macro-action and action policy will be updated respectively (lines 42-43).

Next, we introduce the complexity analysis of Algorithm 1. For outer, middle and inner loops on each episode, the time complexity is $O(M)$, $O(T)$ and $O(K)$ respectively. Based on the conclusion in [43] and [44], the time complexity of actor network is depended on the structure of DNN and the size of state space. Because the number of network nodes is n , the number of neurons in the input layer of actor network is also n . The output of actor network is the parameters of normal distribution with 2 neurons. Meanwhile, we set the number of hidden layer of actor network to q and the number of neurons in each hidden layer to u . Therefore, the total time complexity of Algorithm 1 is $O(MTK(n + qu + 2))$.

VII. PERFORMANCE EVALUATION

To demonstrate the effectiveness of our proposed CM-MTD scheme, we conduct a series of simulations and implement a prototype system. We analyze CM-MTD scheme in terms of prediction, defense, network and convergence performances.

We firstly implemented CM-MTD on wireless network emulator called Mininet-WiFi [45] with version 2.5, and adopted Ryu 4.34 [46] as the SDN controller. In our simulations, the network topology consists of 12/100 OpenFlow switches and 12/100 network nodes. Networkx 2.5 [47] is used to generate the network topology based on Waxman model with parameters $\alpha = 0.2$ and $\beta = 0.15$. According to [48] and [49], some parameter settings have been proven to obtain good training performance. For example, GAE parameters are set as $\gamma = 0.99$ and $\lambda = 0.95$, clip coefficient in PPO is 0.2 and learning optimizer adopts Adam. The activation functions of DQN and PPO are *ReLU* and *tanh* respectively. In addition, we adjust some parameter settings based on the scenario of DTMN. For example, the hidden layers of DQN in the upper layer are set as [256], and the hidden layers of PPO in the lower layer are set as [256, 256]. The simulation parameters are shown in Table I.

As shown in Fig. 6, we also implemented the proof-of-concept of CM-MTD. Our prototype system consists of 4 OpenFlow switches (H3C S5560X-30C-EI), an SDN controller (Intel(R) Core(TM) i9-10920X CPU @ 3.5GHz) with an Nvidia RTX 5000 GPU, and a wireless access point. Raspberry Pis and servers are regarded as end nodes, which connect to OpenFlow switches by wired or wireless manner. The right

TABLE I
SIMULATION PARAMETERS

Parameter	Value or Range
Parameters of Waxman model	$\alpha = 0.2, \beta = 0.15$
Number of OpenFlow switches	[12, 100]
Number of IoT devices	[10, 90]
Number of cloud servers	[2, 10]
Number of IP address spaces	[30, 150]
Communication range of OpenFlow switches	250 m
Link bandwidth	5 Mbps
Size of IP address space	128
Number of episodes in upper layer	10000
Number of steps on each episode	25
Mini-batch size	5
GAE parameters	$\gamma = 0.99, \lambda = 0.95$
Clip coefficient in PPO	0.2
Hidden layers of DQN in the upper layer	[256]
Activation function	<i>ReLU</i>
Learning rate in DQN	10^{-3}
Hidden layers of PPO in the lower layer	[256, 256]
Activation function	<i>tanh</i>
Learning rate in PPO	10^{-4}
Learning optimizer	Adam

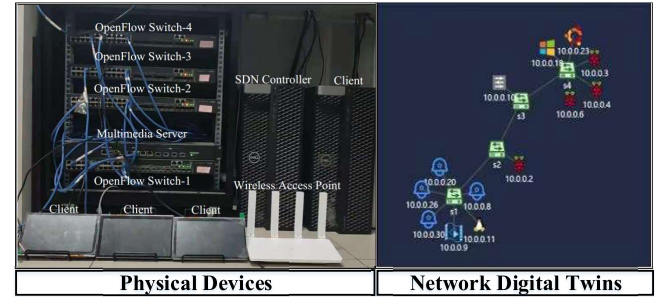


Fig. 6. The proof-of-concept prototype system of CM-MTD.

part in Fig. 6 shows the network scenario in DT space. Our proposed CM-MTD is deployed on the Ryu controller that can be both used in Mininet-WiFi and proof-of-concept prototype system.

A. Prediction Performance

Attack prediction is evaluated based on the event sequence generated by representative attack strategies and real-world dataset called CICIDS-2017 [50]. Attack strategies adopted in event sequence generation include direct DDoS and sequential scanning [51]. CICIDS-2017 contains benign and abnormal network traffic captured within five consecutive days. For direct DDoS, the adversary is more likely to launch DDoS against network nodes with the high node degree. For sequential scanning, the adversary sends probing requests sequentially in scanning space, and will reselect its beginning IP address randomly in a period of time. Therefore, we mainly focus on three types of security events, which are benign, infiltration, and DoS/DDoS. To measure the effectiveness of attack prediction, we adopt prediction accuracy fidelity as the metric, which is shown as follows:

$$Fidelity = \frac{\sum_{i \in N} \mathbb{Y}(p_i, y_i)}{|N|}, \quad (18)$$

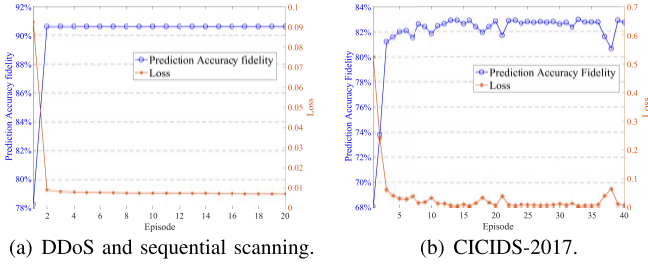


Fig. 7. The prediction accuracy fidelity and loss of attack prediction.

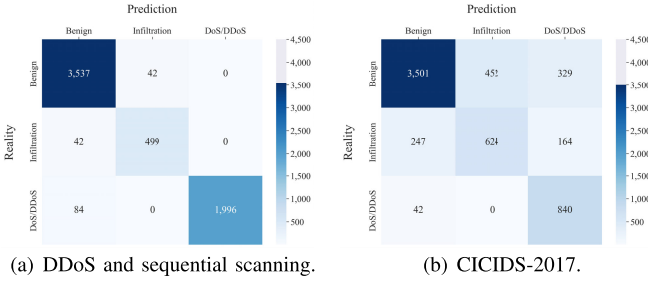


Fig. 8. The confusion matrix on the testing set.

where p_i is the predicted security event of network node v_i , y_i is the actual security event of network node v_i , and $|N|$ is the total number of security events that occur on all network nodes. If the value of $\mathbb{Y}(p_i, y_i)$ is equal to 1, the predicted security event p_i is the same as the actual security event y_i . Otherwise, its value is equal to 0. In addition, we take 80% of dataset as the training set, and 20% of dataset as the testing set.

Fig.7(a) shows the prediction accuracy fidelity and loss of CM-MTD under the event sequence generated by DDoS and sequential scanning. The prediction accuracy fidelity improves from 78% to 92% rapidly while loss decreases from 0.09 to 0.01 within 2 episodes. Under the event sequence generated by CICIDS-2017, simulation results in Fig.7 show that the prediction accuracy fidelity of CM-MTD converges from 68% to 83% within 10 episodes. Meanwhile, the loss of CM-MTD decreases from 0.5 to 0, which indicates that the training has been completed. By comparing simulation results in two event sequences, real-word dataset CICIDS-2017 is difficult to be predicted because the occurrence of malicious behaviors is not always regular.

Furthermore, we evaluate confusion matrices on testing sets under two event sequences respectively, whose results are shown in Fig.8(a) and Fig.8(b). Based on two confusion matrices, we further calculate the prediction accuracy, recall and F1-score of two event sequences. As shown in Fig.8(a), Fig.8(b) and Table II, the prediction accuracy of DoS/DDoS is the highest while the prediction accuracy of infiltration is the lowest. This is because under DDoS+sequential scanning or CICIDS-2017, DoS/DDoS usually lasts for a period of time, which is effective to be predicted. On the contrary, infiltration (e.g., IP scanning) does not happen continuously, and occurs between benign traffic. Therefore, infiltration is difficult to be predicted. With a similar reason, the recall and F1-score of benign are the highest while the recall and F1-score of

TABLE II
SIMULATION RESULTS

Event sequence	Type	Prediction accuracy	Recall	F1-score
DDoS and sequential scanning	Benign	96.56%	98.83%	97.68%
	Infiltration	92.24%	92.24%	92.24%
	DoS/DDoS	100%	95.56%	97.94%
CICIDS-2017	Benign	83.33%	92.37%	87.62%
	Infiltration	60.29%	57.99%	59.12%
	DoS/DDoS	95.24%	63.02%	75.85%

infiltration are the lowest. Simulation results show that our proposed CM-MTD can predict future security events with high accuracy.

B. Defense Performance

In this section, we also evaluate the defense performance by using two event sequences generated by representative attack strategies and a real-world dataset called CICIDS-2017 [50]. As a metric to measure defense performance, defense success ratio (DSR) is denoted as follows:

$$DSR = (1 - \frac{\sum_{k=1}^n N_k^s + \sum_{k=1}^m N_k^d}{\sum_{k=1}^n L_k^s + \sum_{k=1}^m L_k^d}) \times 100\%, \quad (19)$$

where N_k^s is the number of network nodes compromised by scanning, N_k^d is the number of OpenFlow switches compromised by DDoS, L_k^s is the total number of scanned network nodes, and L_k^d is the total number of OpenFlow switches on the transmission route.

Considering that our work is the first intelligent MTD method with the dynamic scheduling and joint deployment of HAM and RM, we compare CM-MTD with following baselines that can be adapted and applied in defending multi-step coordination attacks.

- 1) RRT [31]+FRVM [28], [29]: RRT executes the RM with adaptive mutation period, which is suitable for defending against DDoS. FRVM converts real IP addresses to random and time-varying virtual IP addresses, which is designed for network scanning. As a baseline, we deploy RRT and FRVM simultaneously.
- 2) DQ-RM [32]+FRVM [28], [29]: DQ-RM is the state-of-the-art RM method that dynamically adjusts routes based on DRL. As another baseline, we deploy DQ-RM and FRVM simultaneously.

To evaluate the defense performance effectively, we generate two attack sequences by using direct DDoS + sequential scanning [51] and crossfire DDoS + sequential scanning [52]. For direct DDoS, the adversary is more likely to launch DDoS against network nodes with high degree. For crossfire DDoS, the adversary firstly selects a targeted area, and then sends malicious traffic to neighbour nodes, which will throttle shared links and lead to compromised nodes in targeted area indirectly. For sequential scanning, the adversary sends probing requests sequentially in scanning space, and will reselect its beginning IP address randomly in a period of time. We calculate average DSRs on each 1000 episodes while CM-MTD, RRT+FRVM and DQ-RM+FRVM are deployed respectively. Evaluation results in following figures are averaged over

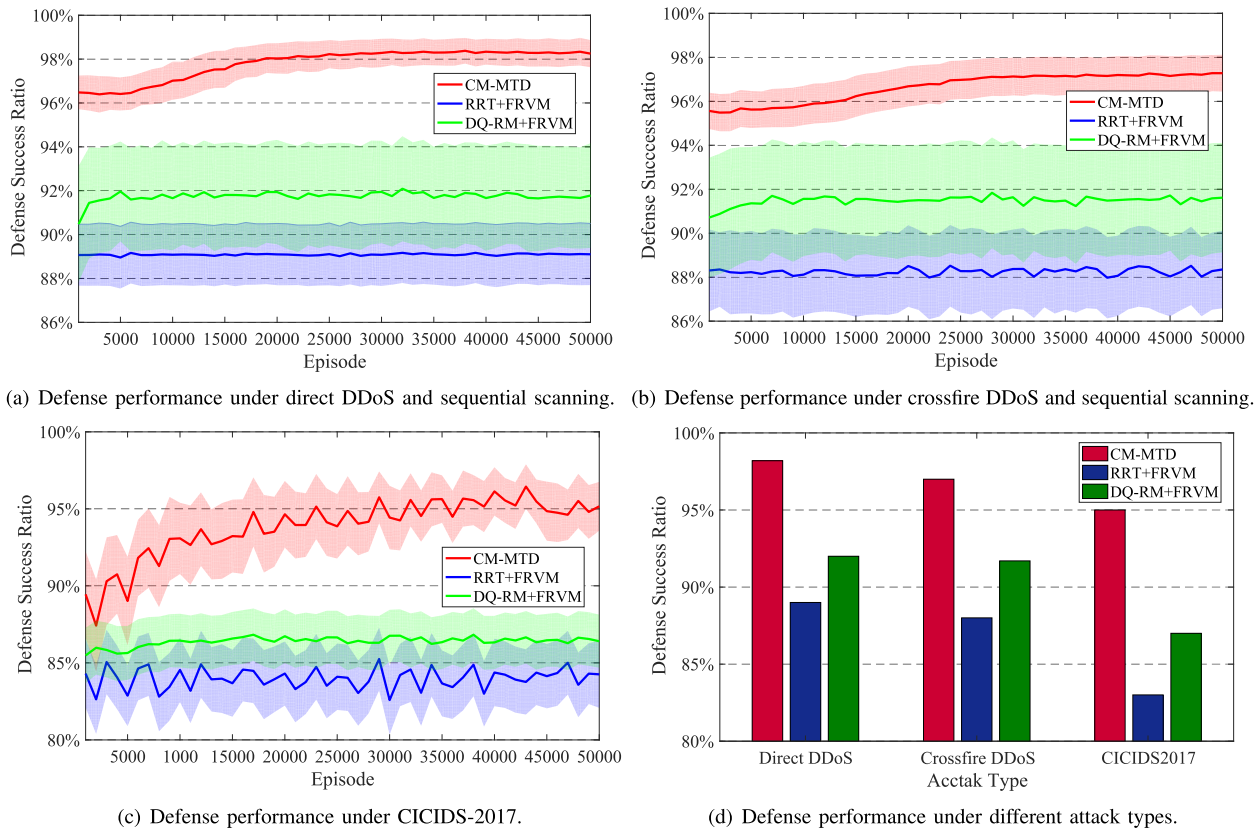


Fig. 9. Defense performance comparison while CM-MTD, RRT+FRVM, and DQ-RM+FRVM are deployed respectively.

5 simulation runs. In addition, the standard errors of evaluation results are indicated as error shadow.

As shown in Fig.9(a) and Fig.9(b), the average DSRs of CM-MTD are the largest among three algorithms, which converge to 98.2% and 97% respectively. The reason is that CM-MTD can optimize the selection of mutated routes and assigned IP address spaces by learning from malicious behaviors. The average DSR of DQ-RM+FRVM converges from 90.5% to 92% because DQ-RM chooses mutated routes based on DQN, and improves defense performance to some extent. On the contrary, the DSR of RRT+FRVM does not change over episodes because it adopts random mutation strategy. Simulation results in Fig.9(c) show that the average DSR of CM-MTD converges from 89% to 95% by learning from attack traces in CICIDS-2017. Because of similar reason in Fig.9(a) and Fig.9(b), the average DSRs of RRT+FRVM and DQ-RM+FRVM are about 83% and 87%, which are smaller than that of CM-MTD. Fig.9 compares defense performance under different attack event sequences. The average convergence DSR of CM-MTD under direct DDoS is 98.2% while the average convergence DSR of CM-MTD under CICIDS-2017 is only 95%. This is because attack traces in CICIDS-2017 is difficult to be learned. As a conclusion, CM-MTD has the best defense performance compared with representative solutions under two attack sequences and a real-world dataset.

C. Network Performance

To evaluate the network performance, we adopt two metrics, i.e., round-trip time (RTT) and packet loss ratio (PLR), which

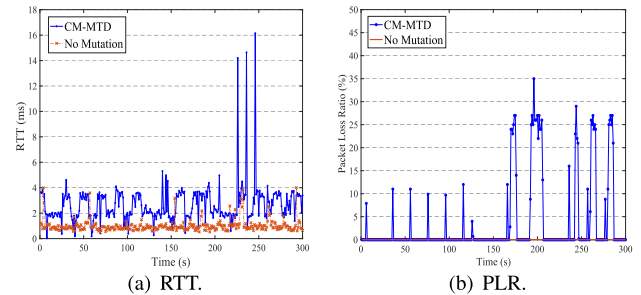


Fig. 10. Network performance comparison between CM-MTD and no mutation.

are measured by iPerf 2.0.10 [53]. RTT is defined as the interval of round transmission time between source and destination. PLR is defined as the portion of packets that are not received by the destinations out of the total packets generated.

We compare the average RTT and PLR with 300 seconds when CM-MTD and no mutation are deployed respectively. The packet sending rate is set as 9Mbits/s. As shown in Fig.10(a), the average RTT of CM-MTD is mostly between 2ms and 4ms while that of no mutation is about 1ms. The reason is that CM-MTD dynamically adjusts mutated routes and/or assigned IP address spaces, which will lead to the sudden increase of RTT. Simulation results in Fig.10 show that the average PLR of CM-MTD is mostly 0%, but it becomes larger sometimes because of RM and HAM. It can be concluded that CM-MTD just affects network performance slightly because CM-MTD adaptively adjusts the deployment time allocation of RM and HAM. Compared with deploying RM and HAM all the time, CM-MTD can reduce their deployment time, which

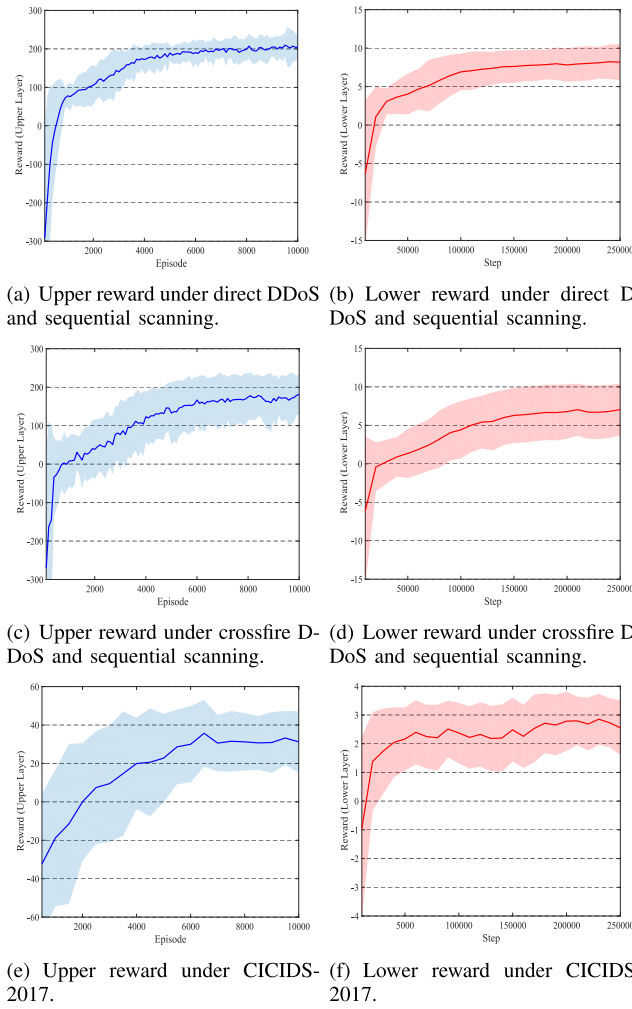


Fig. 11. Convergence performance comparison under different attack sequences.

means that can reduce network overheads caused by MTD execution.

D. Convergence Performance

To evaluate convergence performance, we adopt the rewards of upper and lower layers in the training process of HDRL. Here, the upper layer refers to DQN, and the lower layer refers to PPO. We calculate average reward on each 1000 episodes under two generated attack sequences and a real-world dataset called CICIDS-2017. Evaluation results in following figures are averaged over 5 simulation runs. In addition, the standard errors of evaluation results are indicated as error shadow.

As shown in Fig.11(a), Fig.11(c) and Fig.11(e), the upper layer will converge with about 6000 episodes. Simulation results in Fig.11(b) and Fig.11(d) show that the lower layer converges with about 20000 steps. However, as shown in Fig.11(f), the lower layer only converges with about 5000 steps. This is because attacked targets under CICIDS-2017 are easy to be learned while attacked targets under two generated attack sequences are various based on corresponding attack strategies, which is hard to be learned. In a conclusion, our proposed CM-MTD can converge under two attack sequences and a real-world dataset.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we propose the CM-MTD scheme to collaboratively schedule multiple MTD schemes. Firstly, we model the dynamic deployment of multiple MTD schemes as an SMDP. The future security events of network nodes are predicted by LSTM, which are regarded as the current network state. Next, we formulate feasible mutated routes and assigned IP address spaces based on SMT formalization. Therefore, infeasible actions will be removed from the action space of SMDP. Then, we design a novel HDRL algorithm for collaborative mutation with RM and HAM. Finally, we conduct a series of simulation experiments and implement the proof-of-concept prototype system to confirm the feasibility and effectiveness of CM-MTD.

For the future work, we will investigate how to improve the accuracy of attack prediction. In addition, we will also discuss the reward design to improve the convergence speed of CM-MTD particularly, and make our method adapt to new environment quickly.

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